

Glossary — Cross-Domain Circuit Analysis

Purpose: Plain-language definitions of key concepts used in the paper, for explaining the work to others.

Companion to: CROSS_DOMAIN_CIRCUIT_PAPER.md

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Jaccard Similarity

Quick definition: A number between 0 and 1 that measures how much two sets overlap.

Formula: $J(A, B) = \frac{|A \cap B|}{|A \cup B|} = \frac{\text{size of overlap}}{\text{size of combined set}}$

How to read the values:

- **J = 0** → the two sets share nothing
- **J = 1** → the two sets are identical
- **J = 0.5** → half the elements are shared

Worked example:

Set A = {apple, orange, banana, grape} Set B = {apple, orange, pear, kiwi}

- Intersection: {apple, orange} → size 2
- Union: {apple, orange, banana, grape, pear, kiwi} → size 6
- **J = 2/6 ≈ 0.33**

How we use it: Every circuit has a set of SAE features that activate during the prompt. We compute Jaccard over those feature sets to ask "how similar are these two circuits?"

Example results from the paper:

Comparison	Jaccard	Interpretation
Gold vs Sodium (same template, different facts)	0.60	Circuits share 60% of features
Gold template-A vs Gold template-B (same fact, different templates)	0.30	Only 30% shared
Chemistry vs Geography (different domain)	0.11	Very different circuits

Comparison	Jaccard	Interpretation
GEMMA Geography prompts (within-domain)	0.40	Highest convergence within-domain
GEMMA History prompts (within-domain)	0.11	Lowest convergence within-domain

Why we chose it:

- Simple and interpretable
- Doesn't care about set size — two small sets with perfect overlap still score 1.0
- Ignores element ordering and magnitude

Limitation: Treats all features as equally important. A strongly-activated feature counts the same as one that barely fires.

Activation Energy

Quick definition: A measure of how much "work" a neural network layer is doing, quantified by the magnitudes of feature activations.

How we compute it:

- **Node energy** = sum of activation magnitudes across all SAE features in that layer
- **Total activation energy** = sum of node energies across the entire circuit
- **Layer energy fraction** = (energy at layer L) / (total circuit energy) — what % of the circuit's work happens at layer L

Why call it "energy"? Physics analogy. High energy = many features strongly firing. Low energy = sparse activation. The name captures the intuition that a layer with high activation magnitudes is "doing more work."

How it looks in the paper:

Energy profiles plot layer-energy-fraction against layer depth for each circuit. We found two distinct patterns:

- **GEMMA front-loads energy:** 54% of total energy is in layers 0-12 (first half). Cumulative 50% reached by L11.
- **QWEN back-loads energy:** Only 31% of energy in the first half. Cumulative 50% not reached until L28.

These are fundamentally different computational strategies, but both produce comparable factual recall.

Key property: Energy profiles are **architecture-determined, not domain-determined**.

- Within-model cosine similarity (across chemistry/geography/history): **0.978**
- Between-model cosine similarity: **0.696**
- Architecture explains roughly 14× more variance than knowledge domain

Why this matters: Energy profiles are robust to prompt-template confounds (see Section 5.9). Feature IDs change when you rephrase prompts, but *where* computation happens across layers stays architecturally locked. That's why the architecture-dominance claim survives the format-variation critique.

Bottleneck Tax

Quick definition: The empirical finding that **higher activation magnitude at a bottleneck layer correlates with worse model confidence.** The name is a memorable label; the theoretical grounding is information bottleneck theory (Tishby & Zaslavsky, 2015), which predicts that over-compression at an intermediate layer limits the information downstream layers can use for prediction.

Important note on framing: This is a **correlation with a theoretical interpretation, not a proven causal mechanism**. "Energy" here means summed feature activation magnitudes, NOT a conserved physical quantity (see the "Is Energy Finite?" entry). Early drafts used "energy budget" language; the final paper grounds the finding in information bottleneck theory instead.

The core correlations (GEMMA-2-2B):

Layer	Role	Correlation with confidence	Direction
L6	Primary bottleneck	$r = -0.684, p < 0.0001$	Negative (tax)
L10	Mid-layer	$r = -0.601$	Negative
L1	Input	$r = +0.634$	Positive
L13	Post-bottleneck	$r = +0.601$	Positive
L16	Post-bottleneck	$r = +0.598$	Positive

More energy at L6 → worse predictions. More energy at L13/L16 → better predictions.

The intuition (doorway analogy — use cautiously): Think of a narrow doorway in a hallway.

- Information has to squeeze through the doorway (bottleneck at L6)
- The *useful assembly* happens in the rooms after the doorway (L13, L16)
- When too much is compressed at the doorway, less useful information reaches the downstream rooms for assembly

This is a metaphor to build intuition. The **formal version** comes from information bottleneck theory: the mutual information $I(X; T_L)$ between input X and the representation at layer L bounds what downstream layers can recover. Higher compression at the bottleneck layer reduces this mutual information, which limits downstream evidence accumulation.

Note: The bottleneck both filters information (potentially helpful, by discarding noise) AND limits downstream information (potentially harmful, if it discards signal). The paper's finding is that on average the harm dominates — circuits with heavier bottleneck activation tend to produce less confident predictions.

What we ruled out:

- **Bottleneck convergence** (how sharply paths funnel): $r = 0.27$, NOT significant
- **Bottleneck energy fraction** (what % of energy is at bottleneck): $r = 0.26$, NOT significant
- **Total activation energy**: $r = 0.53$, Bonferroni-significant

So it's not about having a "sharper" or "cleaner" bottleneck — it's about not *over-spending* energy there.

Why this matters: This is one of the paper's most novel contributions because it **connects circuit structure directly to behavior**. Earlier work (Meng et al. 2022) showed specific layers matter for factual recall via causal intervention. Our bottleneck tax is the correlational version: just by measuring *where* energy lives in a circuit, you can predict *how confident* the model will be.

QWEN3-4B contrast: QWEN shows zero Bonferroni-significant layer-energy-confidence correlations. Its late-bottleneck architecture (L22-25) diffuses the confidence signal across many layers instead of localizing it. The bottleneck tax appears specific to early-bottleneck architectures like GEMMA.

One-line summary: Bottleneck tax = an empirical correlation (higher L6 activation → lower confidence) grounded in information bottleneck theory (Tishby 2015), not a literal energy-budget mechanism.

How to explain it to someone: > "We found that the more a circuit's activation is concentrated at its bottleneck layer (L6 in GEMMA), the less confident its prediction tends to be — $r = -0.684$, Bonferroni-significant. This is consistent with information bottleneck theory, which predicts that heavy compression at intermediate layers limits the information downstream layers can use. We call the pattern the 'bottleneck tax' for memorability, but it's a correlational finding with a theoretical interpretation, not a proven causal mechanism."

Cosine Similarity on Energy Profiles

Quick definition: A number between 0 and 1 measuring whether two circuits have the same *shape* of layer-by-layer energy distribution, regardless of total magnitude.

What the inputs are: Each circuit produces a vector of N numbers (26 for GEMMA, 36 for QWEN) — the proportion of total circuit energy at each layer. Example:

GEMMA circuit → [0.08, 0.12, 0.15, 0.09, 0.06, 0.04, 0.03, 0.02, ...]

These sum to 1.0 (they're fractions).

Why cosine specifically:

1. **Shape over magnitude.** Measures the *angle* between vectors, not their length. Two circuits concentrating energy at L6 will score as similar even if one has much more absolute energy. We care about *where* computation happens, not how much.
2. **Standard in ML.** Used ubiquitously for comparing embeddings, feature vectors, and activation patterns.
3. **Interpretable.** 1 = identical shape, 0 = completely different shape.

Alternatives we didn't use and why:

- **Euclidean distance:** dominated by absolute magnitude differences, not shape
- **KL divergence:** asymmetric, treats vectors as probability distributions (which is OK since ours sum to 1, but harder to interpret as a "similarity")
- **Correlation:** adds a mean-centering step that's harder to interpret here

Precedents in the literature:

- **Representational Similarity Analysis (RSA)** — Kriegeskorte, Mur, & Bandettini (2008), *Frontiers in Systems Neuroscience*. Standard for comparing activation patterns across brains/models/conditions.
- **Centered Kernel Alignment (CKA)** — Kornblith, Norouzi, Lee, & Hinton (2019), *ICML*, "Similarity of Neural Network Representations Revisited." Modern method for comparing activations across networks.
- **Cosine on SAE features** — Bricken et al. (2023), Templeton et al. (2024) use cosine similarity on feature activations.

What's novel in our application: Applying cosine similarity specifically to **per-layer energy fractions aggregated across a circuit**, rather than comparing individual activation vectors. It's a reasonable extension of standard practice.

The key results (Section 5.5.1):

- Within-model cosine similarity: **0.978**
- Between-model cosine similarity: **0.696** (after depth normalization)
- Mann-Whitney U test: **$p < 0.000001$** (distributions don't overlap)
- Bootstrap 95% CI on within-model mean: **[0.975, 0.981]**

Where we're vulnerable to reviewer critique:

- We didn't formally benchmark cosine against KL divergence or Earth Mover's Distance. A reviewer could reasonably ask "would the pattern hold with a different metric?"
- The "~14× more variance" claim divides the between-model gap (0.282) by the within-model domain gap (~0.02). This is a rough comparison, not a formal variance decomposition.

How to explain it: > "Cosine similarity measures whether two circuits concentrate their computation at the same layers — we care about the shape of where work happens, not the absolute amount. This extends representational similarity analysis (Kriegeskorte 2008) and modern activation-comparison methods like CKA (Kornblith 2019). Our finding that within-model scores cluster at 0.978 while between-model scores drop to 0.696 is statistically robust (Mann-Whitney $p < 0.000001$)."

Is Energy Finite? (Addressing the Metaphor)

Quick definition: Our paper frames the bottleneck tax as if the model has a "finite energy budget" that must be spent somewhere. **This is a metaphor, not a mechanism.** Neural networks have no conservation law for activation energy.

The physical reality:

- Each layer's activations are the output of learned linear + nonlinear transformations applied independently
- Layer 6 can activate 0 features or 16,000 features without "using up" anything Layer 13 needs
- Activation magnitudes are bounded by input magnitudes and learned weights, not by a global pool

What IS finite in a neural network:

- **Parameters** (fixed capacity)
- **Compute per forward pass** (fixed FLOPs)
- **Input information content** (bounded by prompt)

But these are different from a "shared energy budget."

So what does the bottleneck tax actually capture?

The correlation ($r = -0.684$ between L6 energy fraction and confidence) is **real data**. The *interpretation* as a resource tradeoff is where we're on thinner ground. Three more rigorous ways to frame the same finding:

Option 1: Correlational (safest for publication)

"Layer 6 energy fraction negatively correlates with output confidence across 30 GEMMA circuits. We do not claim a direct causal mechanism; this pattern could reflect upstream prompt difficulty, domain differences, or genuine compression-accumulation tradeoffs."

Option 2: Information-theoretic (stronger, with citation)

Following **Tishby & Zaslavsky (2015)**, "Deep Learning and the Information Bottleneck Principle" (ITW) and **Shwartz-Ziv & Tishby (2017)**, "Opening the Black Box of Deep Neural Networks via Information":

> "Information bottleneck theory predicts that heavy compression at an intermediate layer limits the information available to downstream layers for reconstruction and prediction. The negative correlation between L6 energy fraction and confidence ($r = -0.684$) is consistent with this prediction: circuits that concentrate more processing at the bottleneck may be compressing harder, leaving less information for evidence accumulation in later layers."

This framing uses **information** (which does have a precise mathematical meaning in this context) rather than **energy** (which doesn't, in this context).

Option 3: Economic metaphor (what we currently wrote)

"Heavy processing at the bottleneck consumes resources that could otherwise contribute to confidence." Evocative but not mechanistically grounded.

References that do NOT support "energy is finite":

- **Energy-based models** (LeCun, Hinton): use "energy" as a scalar objective function over configurations — completely different concept
- **Activation atlases** (Olah et al., Distill): use activation magnitudes as features without framing them as finite
- **Hardware energy consumption** (MLPerf, EfficientNet literature): about Joules used by chips, not about activations

Recommendation for publication:

We should soften Section 6.4 to either Option 1 (correlational) or Option 2 (information-theoretic, with citation). The current "energy tax" framing is evocative but a reviewer who pushes on the mechanism will find nothing solid behind it.

How to explain it honestly: > "The paper uses 'energy' as a convenient shorthand for activation magnitude, not as a claim that the model has a literal conserved quantity. The bottleneck tax is a correlation, not a mechanism. The best theoretical grounding comes from information bottleneck theory (Tishby & Zaslavsky 2015), which predicts that compression at intermediate layers reduces information available downstream — consistent with our observation that high L6 energy predicts low confidence."

What this means for the paper:

- The **data** ($r = -0.684$, Bonferroni-significant) stands.
- The **finding** (certain layers' energy predicts confidence) stands.
- The **interpretation** (why) should be softened to information-theoretic language or pure correlational language.
- The name "bottleneck tax" is fine as a memorable label, but the framing around it should acknowledge it's a metaphor.

How to Extend This Document

When a new question comes up:

1. Add entry to the Table of Contents
2. Follow the structure: Quick definition → Details → Example → Why it matters
3. Keep plain-language definitions at the top of each section so readers who only want the gist don't have to read the full explanation
4. Cross-reference with specific sections of `CROSS_DOMAIN_CIRCUIT_PAPER.md` where relevant